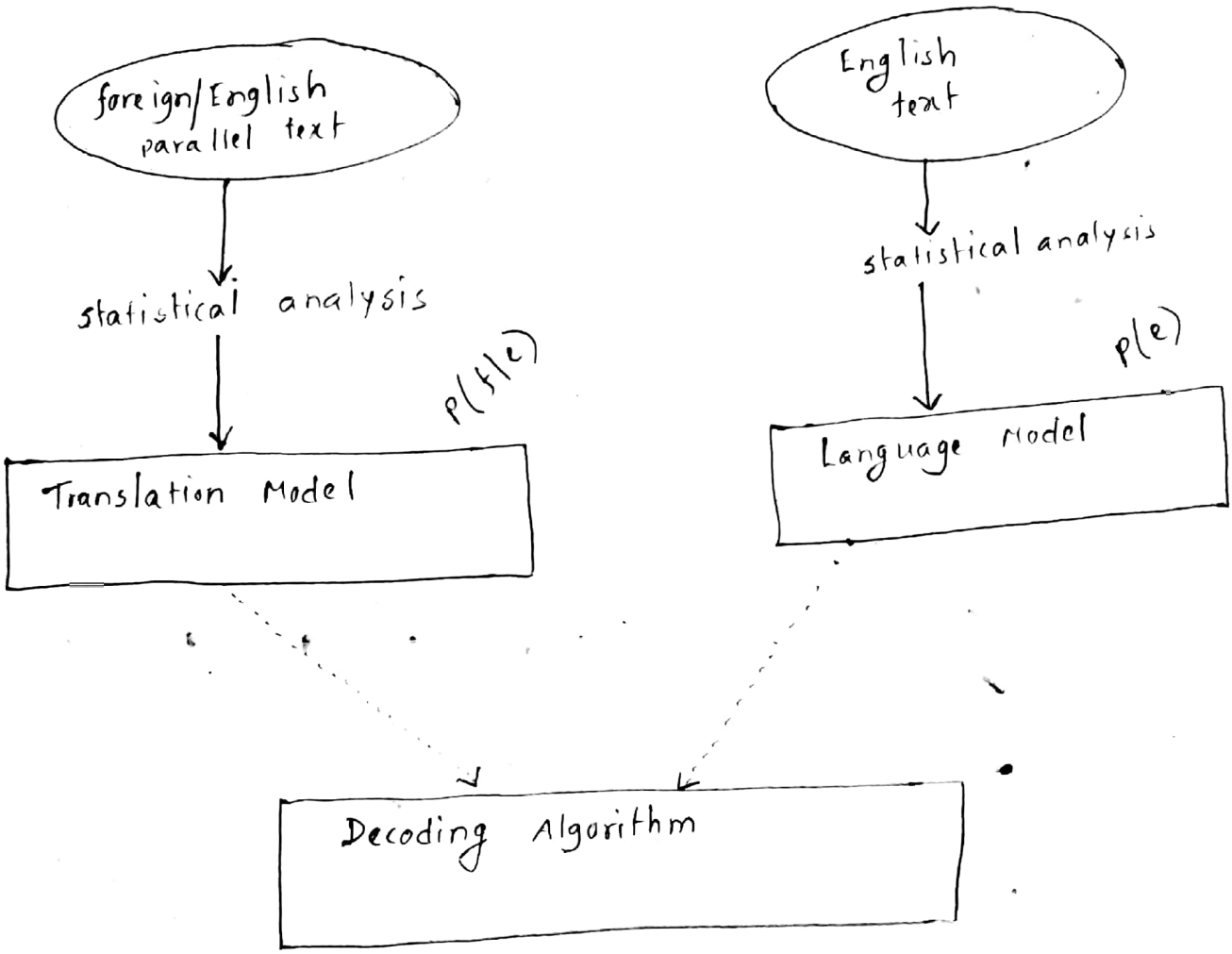
Statistical Machine Translation:

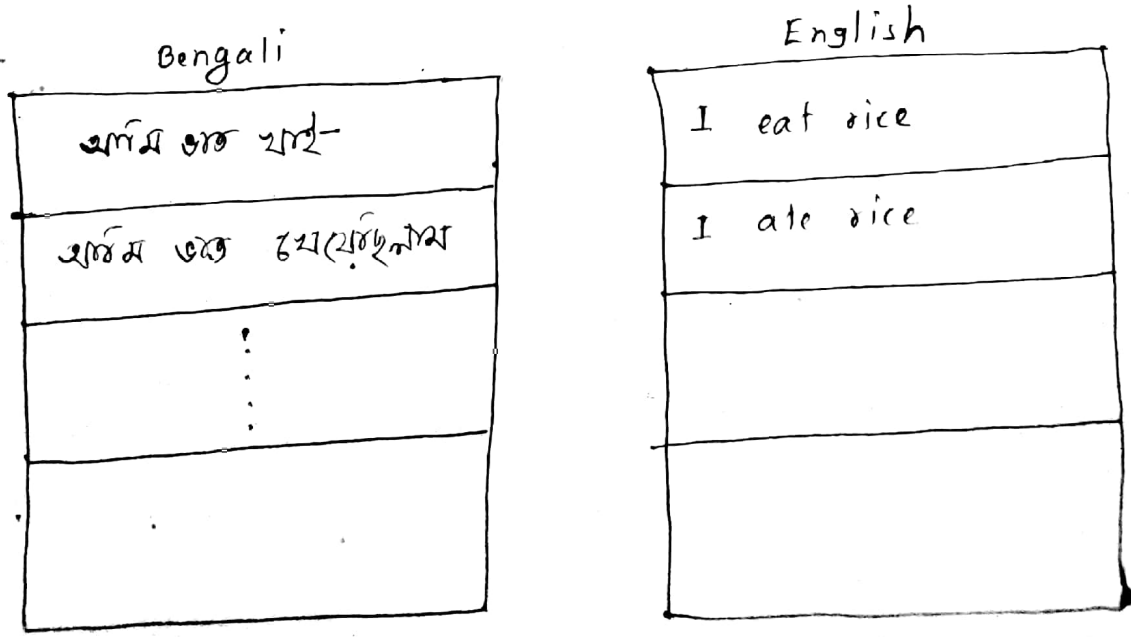
$$P(e|f) = \max\left(P(f|e) * P(e)\right)$$

$P(e)$ → Language model

$P(f|e)$ → Translation model



Parallel Corpus:



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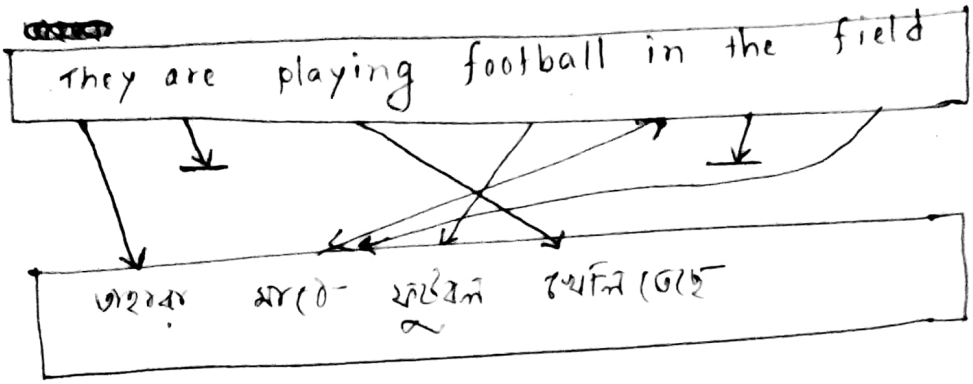
Different Models in statistical Machine Translation

- Word based models.
- Phrase based models.
- Sytax based models.

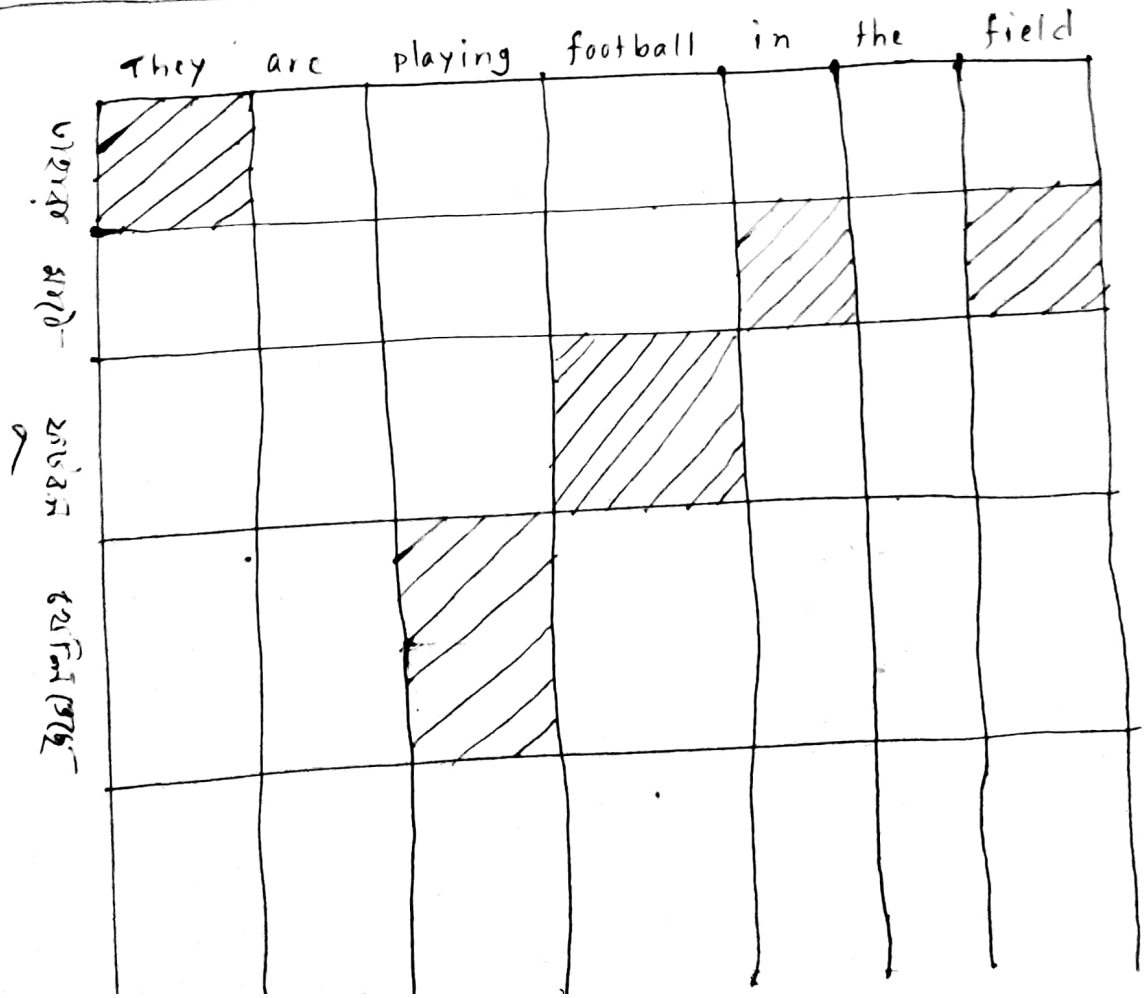
Word based models:

Translation process is decomposed into smaller steps, each is tied words.

Example :



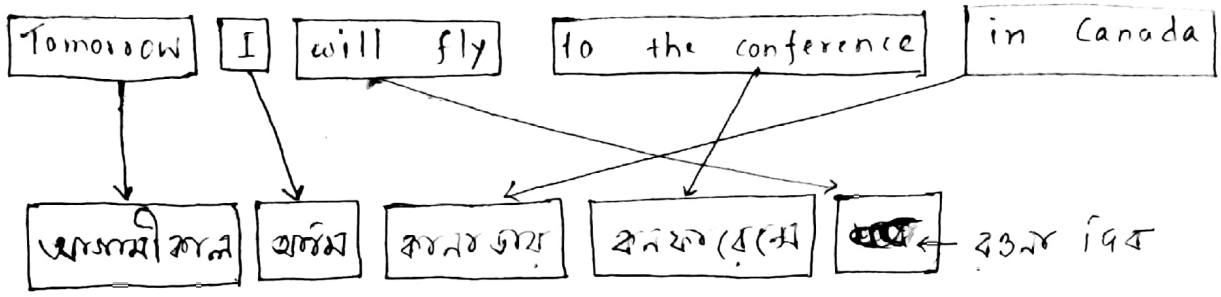
Word alignment:



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phrase-based models:

Ex:



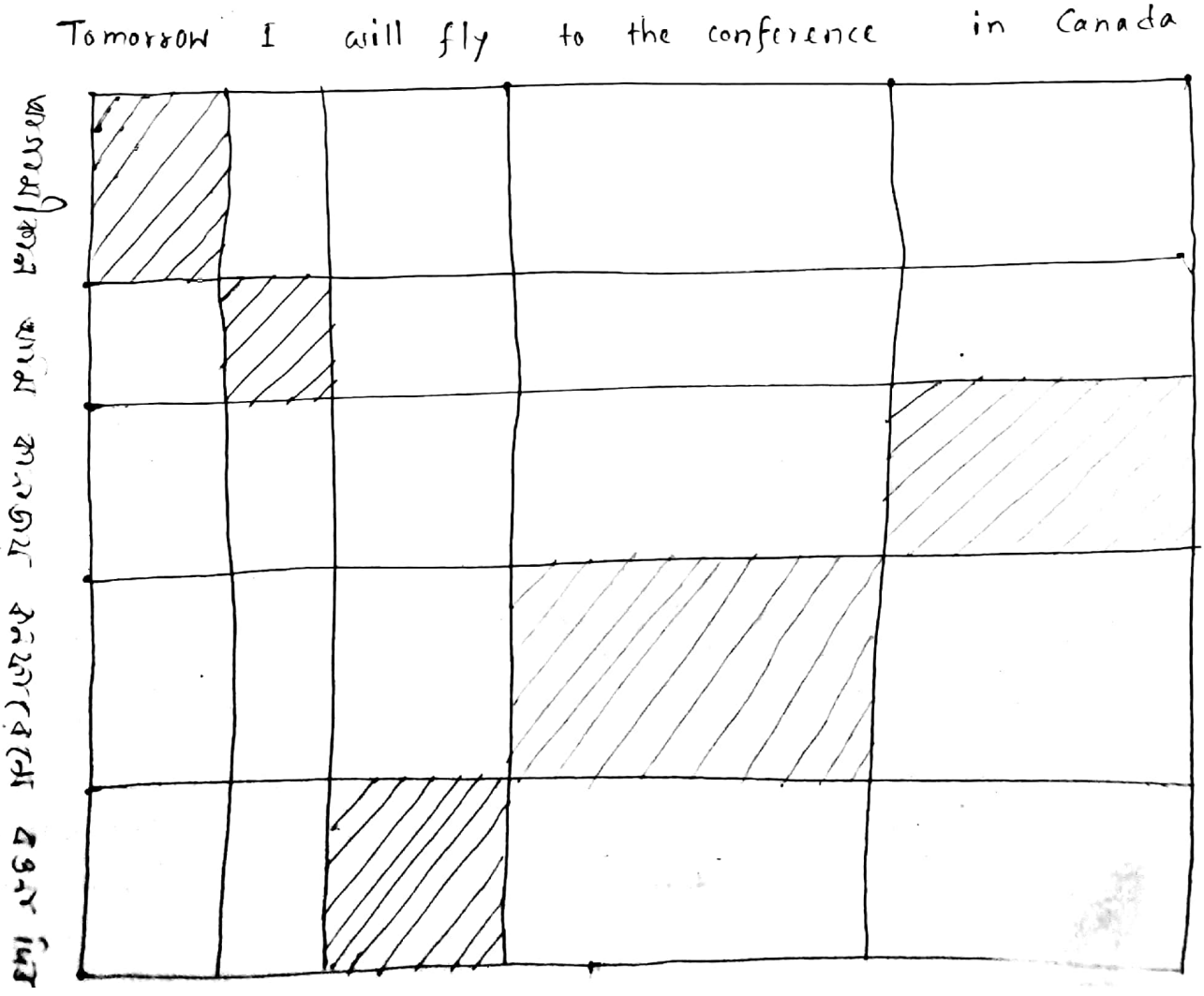
Algorithm:

step1: Foreign input is segmented into phrases
- Any sequence of words

step2: Each phrase is translated into English

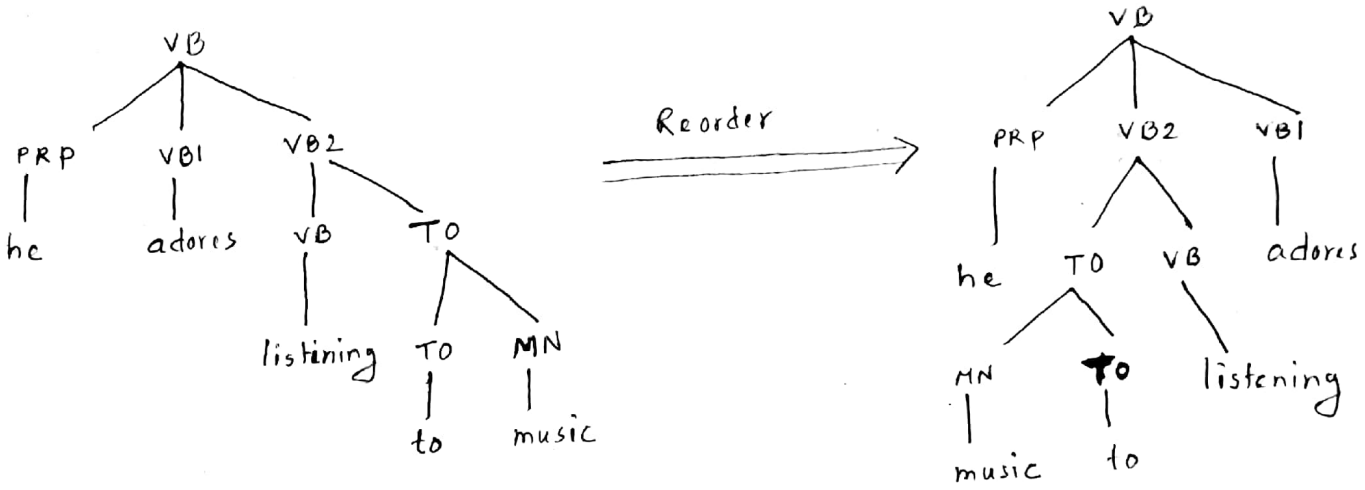
step3: phrases are reordered.

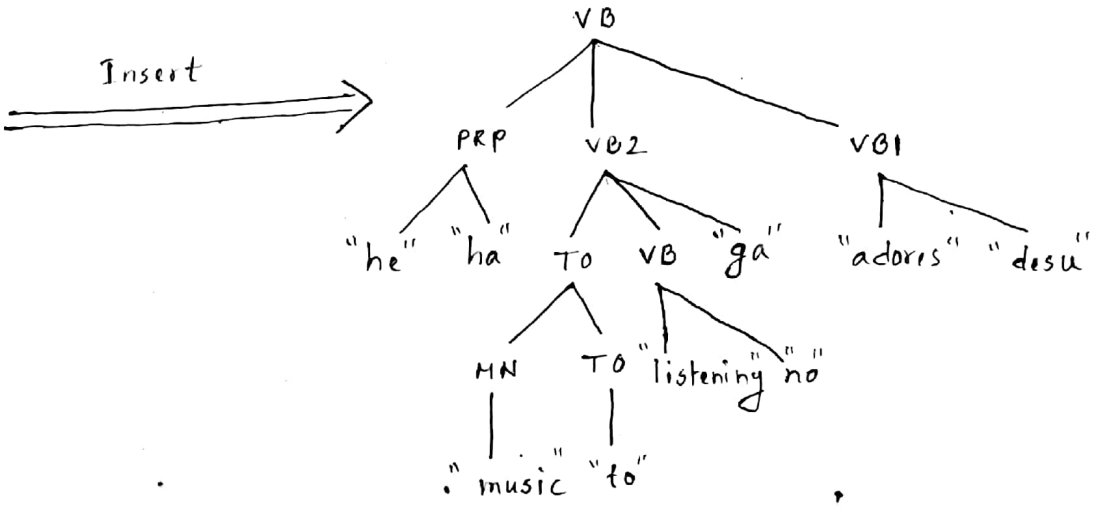
phrase alignment:

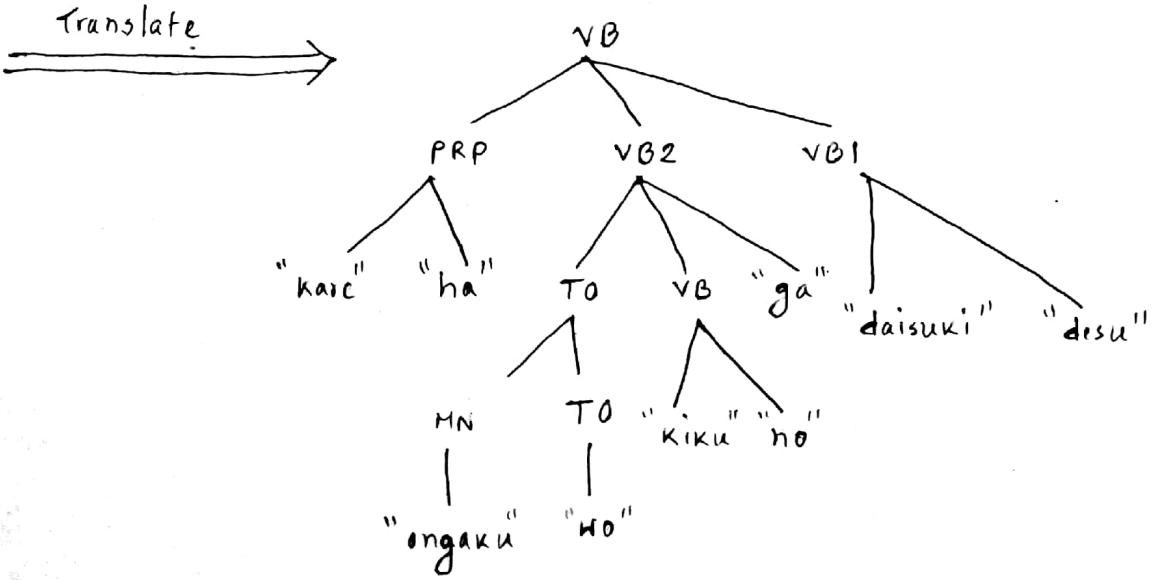


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Syntax-based models:







Take leaves

Kare ha ongaku wo kiku no ga daisuki desu

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Performance :

- Word Error Rate
- Word Inaccuracy

Example:

Bangali: তারা মাঠে ফুটবল খেলছে

English: They are playing football in the field

তারা মাঠে ফুটবল খেলছে

Input: তারা | | খেলছে | ফুটবল | | | মাঠে

Output: They | are | playing | football | | the | in field

Total words, $N = 4$
Substitution, $S = 0$
Deletion, $D = 0$
Insertion, $I = 2$

$$\text{Word Error Rate} = 100\% - \text{Word Correct Rate}$$

$$= 100\% - \frac{N - S - D}{N} * 100\%$$

$$= 100\% - \frac{4 - 0 - 0}{4} * 100\%$$

$$= 100\% - 100\%$$

$$= 0\%$$

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